

Money-in-the-Utility Function (MIUF) Model

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Introduction

- In RBC model, all variables are measured in real terms
- We now introduce money in the model; two common ways to do it
 - CIA: Cash in advance (Clower, 1967)
 - Assumption: Money needs to be used to buy consumption goods
 - Rationale: Nothing in the model explains why money is used
 - Modeling implication: Additional constraint
 - MIUF: Money in the utility function (Sidrauski, 1967)
 - Assumption: Money provides some service to the economy
 - Rationale: Save time, insurance against shocks, reduce search costs
 - Modeling implication: Additional argument in utility function
- **Main finding:** Money is neutral in these models

Assumptions

- Same assumptions about economy as in RBC model
- Same assumptions for HHs and firms as before, except for utility function
- Utility function in period t is of the form

$$U(C_t, \frac{M_t}{p_t}, H_t)$$

where M_t/p_t are real money balances, $U_M > 0$ and $U_{MM} < 0$

- $U(\cdot)$ uses logarithms
- New parameters: D for real money balances and h_0 for labor
- Additional shock: Growth rate of money supply
- Model and handout have same structure as before

Technical: Maximum Cardinality Matching

- Maximum cardinality matching refers to largest set of equations that can be solved simultaneously without creating a dependency loop
- A variable in an equation not in maximum cardinality matching means variable cannot be solved directly in that equation
- Reasons: Circular dependency, variable not directly linked to solved eqs.
- Troubleshooting:
 - Make sure variable not part of a circular dependency
 - Simplify model or re-express equations
 - Ignore error if only interested in simulations or IRFs
 - Relevant error if doing a block decomposition of the model

Technical: Collinearity

- Collinear relationships error
 - Redundant equation(s)
 - (Near) unit roots
 - SS is not unique
 - Jacobian of static model is singular (i.e. no unique solution)
 - They may not affect real variables
- Troubleshooting:
 - Identify and replace redundant equation(s)
 - Simplify model (start with a basic version and gradually add complexity)
 - Provide SS analytically